Simulation of gradient-driven degradation in high-temperature heating elements

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An alumina-forming alloy is protected by an alumina scale that consumes aluminum as it grows, and higher temperatures can accelerate this consumption. This study calculated the temperature field in an extruded honeycomb heater and the time for aluminum to fall to an assumed threshold, called the depletion time. At fixed power, with uniform channel cooling and an insulated outer surface, metal temperatures ranged from 1052.4 to 1115.8 °C, and the outer corners were hottest. Near the outer surface, the parabolic rate constant averaged 1.59–2.03 times the interior value for activation energies of 230 and 360 kJ mol^{-1}. Two checks supported the simplified depletion model: available aluminum changed by less than 0.02% over the calculated temperature range, and the surface reached the threshold less than 1% earlier than the wall mean. Thinning the outer wall to 0.698 of its thickness lowered the maximum temperature by 26.1 K but shortened depletion time to 0.71–0.88 of the unthinned value, because the thinner wall held less aluminum. Exterior convection up to 20 W m^{-2} K^{-1} did not change this. For the insulated walls, thinning would lengthen depletion time only if the activation energy exceeded 429–434 kJ mol^{-1}, above both tested values.

I. INTRODUCTION

ELECTRIC resistance heating is established in industrial furnaces used in ceramic, glass and steel processing[1, pp. 5–6]. Industrial heat electrification[2] motivates compact devices that generate heat within the process. Wismann et al.[3] demonstrated this approach for catalytic structures. Sintex's HoneyComb heats process gas by electrical resistance in a channeled body made by metal powder extrusion[4]. The body is made of an alumina-forming alloy, which is protected by an alumina scale.

This scale consumes aluminum as it grows, and regrowth after spallation consumes more[5]. Local temperature increases can accelerate oxide growth and aluminum consumption, so uneven temperatures matter for degradation[5,12]. Surface shape, metal volume per exposed area and local scale failure also affect oxidation-limited lifetime in published ODS FeCrAl specimens[6].

This study examined how the heater temperature field affects the time for aluminum to reach an assumed threshold. This threshold represents insufficient aluminum to maintain protective alumina in the model, rather than a measured failure point. The outer wall is cooled by channels on one side only, so it is likely to be hotter than the interior walls. A thinner outer wall offers a shorter conduction path to the channels. Thinning it therefore tested whether this cooling could compensate for the aluminum removed.

II. METHODS

A. *Model chain, inputs and scenarios*

Four calculations linked the temperature field of the heater to aluminum depletion (Fig. 1b). The thermal model calculated metal temperature $T(x, y)$, where x and y locate points in the outlet cross-section. Equilibrium calculations gave the available aluminum C_0, the aluminum not bound in AlN or oxides, per unit mass of alloy. The temperature and available aluminum were combined with a parabolic oxidation law to calculate a local depletion time. This time assumes uniform aluminum content through each wall, so a diffusion calculation checked this well-mixed wall assumption.

The reference case had an insulated exterior, and the thermal cases varied conductivity, channel convection and outer wall thickness. Insulation gives the highest corner temperatures the model allows, because any exterior loss can only lower them. Additional cases used exterior heat transfer coefficients of 5, 10 and 20 W m^{-2} K^{-1}, with the sink at the gas temperature.

Table I lists the reference inputs and the cases used to compare them. The equilibrium cases used the alloy certificate midpoints for the nominal composition, and its lower and upper aluminum limits for Low Al and High Al.

The alloy composition was used in the calculations, but it and the absolute depletion times are confidential to Sintex A/S, so depletion times are reported relative to a reference case. The temperature field, the rate ratios and the thinning comparisons do not depend on the composition.

B. *Temperature field*

The cross-section was reconstructed from a drawing (Fig. 1a). The outlet gas temperature and channel heat transfer coefficient came from the supplied one-dimensional model. A consistency run used the same wetted perimeter and conductivity as the two-dimensional model. Its outlet values differed by less than 1 K and 1% from the adopted inputs, so these inputs were retained. The channel conditions were applied uniformly, with no iteration between the models.

Stationary conduction was solved with uniform heat generation over the metal. The total power P, element length L and metal area A_s set the heat generated per unit volume, q_v. With conductivity k_s[10, Table 2] and metal temperature T, the equation solved in COMSOL Multiphysics 6.4[9] is

$$\nabla \cdot (k_s \nabla T) + q_v = 0, \qquad q_v = \frac{P}{L A_s}. \tag{1}$$

The model omitted current redistribution, axial conduction and a separate radiation term. A constant MA956-based conductivity served as a surrogate; the conductivity scan tested sensitivity to this choice.

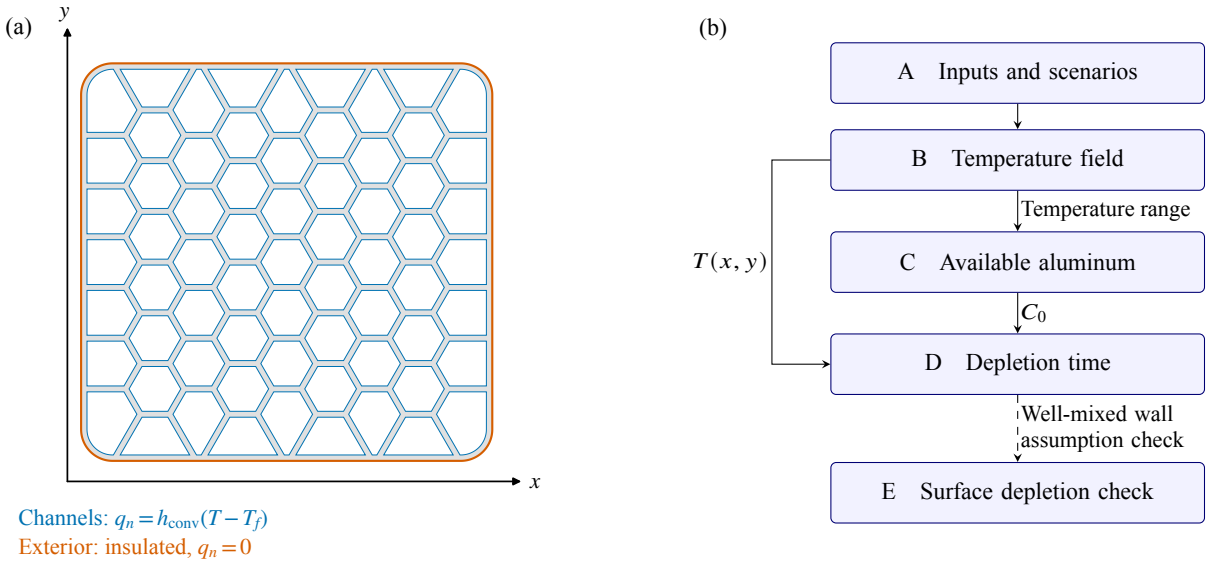


Fig. 1—(a) Reconstructed cross-section and reference thermal boundaries; the coordinate arrows indicate orientation only. The outward heat flux is q_n; h_{conv} is the channel heat transfer coefficient, and T_f is gas temperature. (b) Model chain. The temperature range guides equilibrium sampling; the depletion calculation uses a common initial C_0 and the local temperature. The dashed link denotes the diffusion check of the well-mixed wall assumption.

Table I. Inputs, sources and calculated cases.

Input	Reference value	Source	Varied as
Outlet gas temperature T_f	631 °C	Sintex 1D model	—
Channel heat transfer coefficient h_{conv}	62.1 W m^{-2} K^{-1}	Sintex 1D model	50, 75 W m^{-2} K^{-1}
Conductivity k_s	25 W m^{-1} K^{-1}	Approximation to MA956 near 1000 °C[10, Table 2]	18, 21 W m^{-1} K^{-1}
Outer wall thickness d	t_0	Reference geometry	$0.814t_0$, $0.698t_0$
Exterior heat transfer coefficient h_{ext}	0 (insulated)	Assumed	5, 10, 20 W m^{-2} K^{-1}
Parabolic rate constant k_p at 1100 °C	9.6×10^{-13} g^2 cm^{-4} s^{-1}	Issartel et al.[8], Table 2	×10
Activation energy Q	230, 360 kJ mol^{-1}	Wright and Pint[12], Table II	Both used
Aluminum depletion threshold C_B	2.0 wt%	Assumed, within inferred values[6]	—

With **n** the outward unit normal, the channel condition is

$$-k_s \nabla T \cdot \mathbf{n} = h_{conv}(T - T_f), \tag{2}$$

with the same form used for exterior convection through its heat transfer coefficient h_{ext}.

Integrating the heat balance over the cross-section gives the mean channel-wall temperature $\overline{T}_{cw}$. Here P_{wet} is the channel perimeter, Γ_{ch} and Γ_{ext} are the channel and exterior boundaries, and $q'_{ext} = \int_{\Gamma_{ext}} h_{ext}(T - T_f)\,\mathrm{d}s$ is the exterior heat loss per unit element length:

$$\begin{aligned} \overline{T}_{cw} &= \frac{1}{P_{wet}} \int_{\Gamma_{ch}} T\,\mathrm{d}s \\ &= T_f + \frac{P/L - q'_{ext}}{h_{conv} P_{wet}}. \end{aligned} \tag{3}$$

Each thermal scan changed one independent input while the others stayed at their reference values. The thinner cases kept the outer envelope and internal walls, and q_v was recalculated to keep total power fixed.

The reference mesh had 2862 quadratic elements. Refining it to 8976 elements changed the maximum temperature by +0.18 K and the minimum by −0.14 K. Boundary heat loss matched the generated heat to within 3×10^{-10} in relative terms.

C. *Available aluminum*

Equilibrium calculations determined which phases contained aluminum over the calculated metal-temperature range. Thermo-Calc 2025b with TCFE14 calculated these phases at 10^5 Pa[11]. The calculation used fixed bulk inventories in a closed system. The 500–1500 K range covered all metal temperatures in the thermal cases, up to 1498 K.

Aluminum bound in AlN and oxides was treated as unavailable to the scale and excluded from the initial reserve. Exchange between bound and dissolved aluminum during service was not modeled. A single value of available aluminum at 1350 K was used because its relative variation was below 0.02% (Section III.B). Temperature-dependent partitioning into internal phases could otherwise change available aluminum between walls, preventing its cancellation in depletion-time ratios.

D. *Depletion time*

The oxidation law converted each metal temperature into a parabolic rate constant relative to the reference temperature. The reference rate was measured over 100 h in air at 1100 °C on 1 mm cast and forged specimens of a model FeCrAl with about 19.7 wt% Cr, 5.0 wt% Al and yttrium[8]. The relative oxidation rate g is defined here as the ratio of parabolic rate constants k_p. In the temperature dependence, Q is activation energy and $R = 8.314$ J mol^{-1} K^{-1} is the gas constant;

temperatures are in kelvin:

$$g(T) = \frac{k_p(T)}{k_p(1373.15\ \mathrm{K})} = \exp\left[-\frac{Q}{R}\left(\frac{1}{T} - \frac{1}{1373.15\ \mathrm{K}}\right)\right]. \quad (4)$$

The activation energies were rounded from values reported for Stage 2 air oxidation of MA956HT and MA956[12]. Stage 2 denotes parabolic scale growth with negligible spallation[12, p. 5]. Both laws used the same parabolic rate constant at 1100 °C, which lies within the calculated metal temperatures.

The depletion calculation balanced aluminum in a flat wall against aluminum consumed in the scale. Each plate had full thickness d and bulk density ρ; the initial available aluminum C_0 and threshold C_B are mass fractions. All walls, including the outer wall, were taken to oxidize on both faces; for the outer wall this gives the shorter depletion time. Pores were taken as closed, so the oxidizing area equals the geometric area. All walls had the same density and initial available aluminum.

Aluminum loss was calculated from the oxygen mass gain Δm_O per geometric area A of one oxidizing face. Alloy mass was held fixed during depletion, and all oxygen uptake was assigned to alumina. The factor η converts oxygen mass gain to aluminum consumption. The depletion time t_B is the time at which aluminum reaches the threshold. The mass balance and parabolic uptake give

$$\rho d(C_0 - C_B) = 2\eta\,\frac{\Delta m_\mathrm{O}}{A}, \qquad \frac{\Delta m_\mathrm{O}}{A} = \sqrt{k_p t_B},$$

$$t_B = \frac{1}{k_p(T)}\left[\frac{\rho d(C_0 - C_B)}{2\eta}\right]^2, \qquad \eta = \frac{54}{48}. \quad (5)$$

Equation 5 follows the aluminum balance used by Pint et al.[5], Eq. 1.

The reference time t_ref is for the nominal composition, the reference wall thickness t_0, 1100 °C and $C_B = 2.0$ wt%. With a rate multiplier s, the relative depletion time is

$$\frac{t_B}{t_\mathrm{ref}} = \left(\frac{d}{t_0}\right)^2 \frac{1}{s g(T)}, \qquad s = 1 \text{ or } 10.$$

In cyclic tests on MA956 at 1250 °C, repeated cooling and spallation changed approximately parabolic growth to nearly linear growth[6]. The tenfold rate tested faster consumption. A fixed threshold and a constant rate multiplier do not reproduce that changing growth law.

The outer band is the metal within one reference wall thickness of the outer surface, and the remaining metal is the interior. In each region, g was averaged over metal area, with mesh triangles assigned to a region by their centroids. The ratio of the two means compares the outer band with the interior.

Each thinner outer wall was compared with the reference outer wall at the same exterior heat transfer coefficient. The maximum metal temperatures of the respective cross-sections are T_d for thickness d and T_0 for reference thickness t_0. In every compared case, this maximum lay in the outer band. The ratio of depletion times is

$$\frac{t_B(d, T_d)}{t_B(t_0, T_0)} = \left(\frac{d}{t_0}\right)^2 \exp\left[\frac{Q}{R}\left(\frac{1}{T_d} - \frac{1}{T_0}\right)\right]. \quad (6)$$

Setting this ratio to unity gives the activation energy at which the thinner and reference walls have equal depletion times. Composition, density, threshold and rate prefactor cancel in this ratio.

E. *Surface depletion check*

The diffusion calculation checked the well-mixed wall assumption against surface depletion. Equation 5 tracks mean aluminum, but the surface can reach the threshold earlier if diffusion cannot keep up. DICTRA with TCFE14 and MOBFE8[11] calculated single-phase matrix diffusion at 1100 °C. The domain covered half of the reference wall thickness, from the oxidizing surface to a closed symmetry plane. The ternary, single-phase approximation isolated matrix diffusion; precipitate exchange and pore-scale transport were not resolved.

Aluminum was removed at the oxidizing surface at the rate required by parabolic alumina growth. The positive outward aluminum molar flux is $J_\mathrm{Al}^\mathrm{out}$, and t is elapsed time. Oxygen molar mass is $M_\mathrm{O} = 15.999\ \mathrm{g\,mol^{-1}}$. The time offset τ removed the initial singularity in the imposed flux:

$$J_\mathrm{Al}^\mathrm{out}(t) = \frac{2\times 10^4}{3M_\mathrm{O}}\,\frac{\sqrt{k_p}}{2\sqrt{t+\tau}}, \qquad \tau = 1\ \mathrm{s}. \quad (7)$$

The factor $2/3$ follows Al_2O_3 stoichiometry, and 10^4 converts area units so that the flux is in $\mathrm{mol\,m^{-2}\,s^{-1}}$. No other element left the surface.

The case with ten times the oxidation rate tested the larger difference between surface and mean aluminum content at higher imposed flux. The acceptance criterion was a difference below 5% between surface and mean depletion times, measured relative to the mean depletion time. Surface and mean depletion times were interpolated linearly between saved outputs. The aluminum removed from the wall agreed with the integrated imposed flux to within 0.72%.

III. RESULTS

A. *Thermal field and parameter response*

The hottest metal lay at the outer corner arcs (Fig. 2a). Temperatures ranged from 1052.4 to 1115.8 °C, a difference of 63.4 K, and the mean channel-wall temperature was 1075.6 °C. Raising the conductivity from 18 to 25 $\mathrm{W\,m^{-1}\,K^{-1}}$ lowered the maximum from 1127.7 to 1115.8 °C and the temperature difference from 80.6 to 63.4 K (Fig. 3a). The mean channel-wall temperature did not change. Raising the channel heat transfer coefficient from 50 to 75 $\mathrm{W\,m^{-2}\,K^{-1}}$ lowered the maximum from 1224.9 to 1037.8 °C, but the temperature difference changed only from 66.6 to 60.4 K (Fig. 3b). Thinning the outer wall to $0.698t_0$ lowered the maximum to 1089.7 °C and the temperature difference to 28.7 K (Fig. 3c).

B. *Equilibrium phases and available aluminum*

The matrix remained ferritic across the temperature range of the reference cross-section, while the amounts of carbides and oxides changed (Fig. 4a). AlN was present throughout this range. These phase changes had little effect on the available aluminum. Relative to its value at 1350 K, it changed

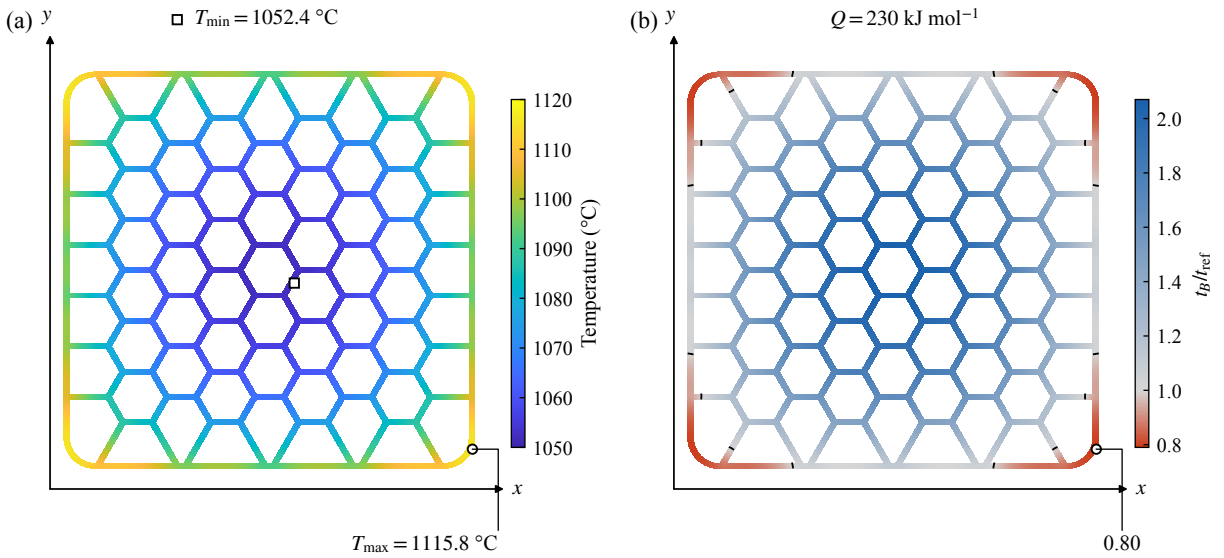


Fig. 2—Reference cross-section: (a) metal temperature; (b) local depletion time relative to 1100 °C at $Q = 230$ kJ mol^{-1} and common thickness and exposure. In (b), black segments mark 1100 °C; the circle gives the hottest corner's relative depletion time. T_{max} and T_{min} are the maximum and minimum metal temperatures.

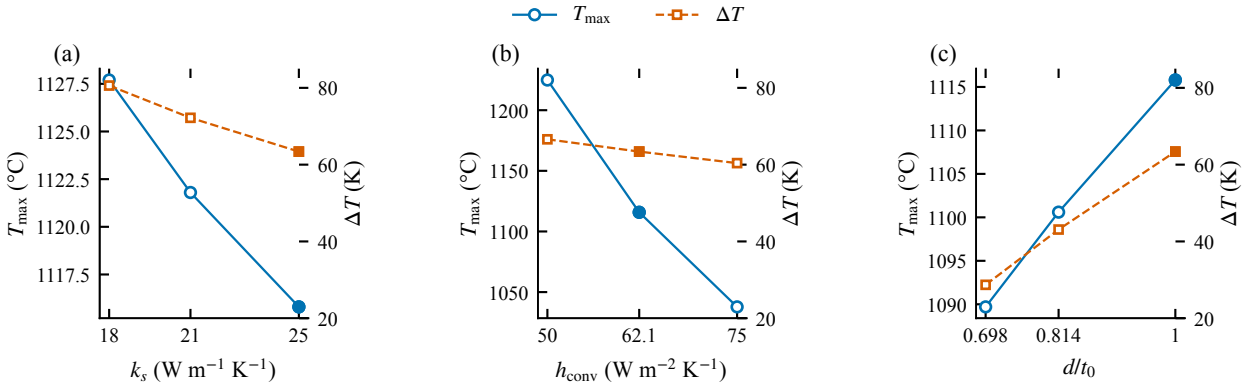


Fig. 3—Scans of (a) conductivity, (b) channel heat transfer coefficient and (c) outer wall thickness d, relative to the reference thickness t_0. Circles show the maximum metal temperature T_{max}. Squares show the difference between maximum and minimum metal temperatures, ΔT. Filled markers identify the reference case.

by less than 0.02% over 500–1500 K in all three composition cases (Fig. 4b).

C. *Oxidation rate ratios and depletion*

The hotter outer band had a larger mean parabolic rate constant than the interior. At an activation energy of 230 kJ mol^{-1}, its mean was 1.59 times the interior value. At 360 kJ mol^{-1}, the ratio was 2.03.

The calculated temperature field gave both shorter and longer depletion times than the reference at 1100 °C. Figure 2b shows this temperature effect on relative depletion time for a plate with common thickness and two oxidizing faces. At the hottest outer corner, the relative depletion time was 0.80 and 0.70 for the two activation energies, respectively. At the coolest point, the corresponding values were 2.06 and 3.10. The assumed tenfold k_p tested faster consumption: it divided depletion times by ten without changing the thinning ratios.

The thinner outer wall contained less aluminum, and its lower temperature did not compensate for this loss. For the insulated wall at $0.698t_0$, cooling alone would have increased depletion time by factors of 1.46 and 1.82 at 230 and 360 kJ mol^{-1}. The reduction in thickness alone reduced depletion time to 0.487 of the reference because time scales with thickness squared. Together, these effects reduced depletion time to 0.713 and 0.884 of the reference wall at the same exterior coefficient.

Equal depletion times for the insulated walls would require 429 kJ mol^{-1} at $0.814t_0$ and 434 kJ mol^{-1} at $0.698t_0$, above both tested values. Higher activation energy amplifies the oxidation-rate reduction from cooling.

Exterior convection lowered the maximum temperature of all three wall thicknesses without reversing their depletion-time ordering. At an exterior coefficient of 20 W m^{-2} K^{-1}, heat lost through the exterior was 4.34–4.53% of input power. Both thinner walls remained below unity at both activation energies for all four exterior coefficients; Table II lists the 0 and 20 W m^{-2} K^{-1} cases.

The diffusion calculation showed a decreasing difference in aluminum content between the wall surface and the wall center (Fig. 5). With ten times the parabolic rate constant, the difference fell from 0.498 percentage points at 1.1 h to 0.063 at 110 h. This narrowing is consistent with the declining imposed flux and diffusion extending across the half wall. The surface reached the threshold 0.108% and 0.992% earlier

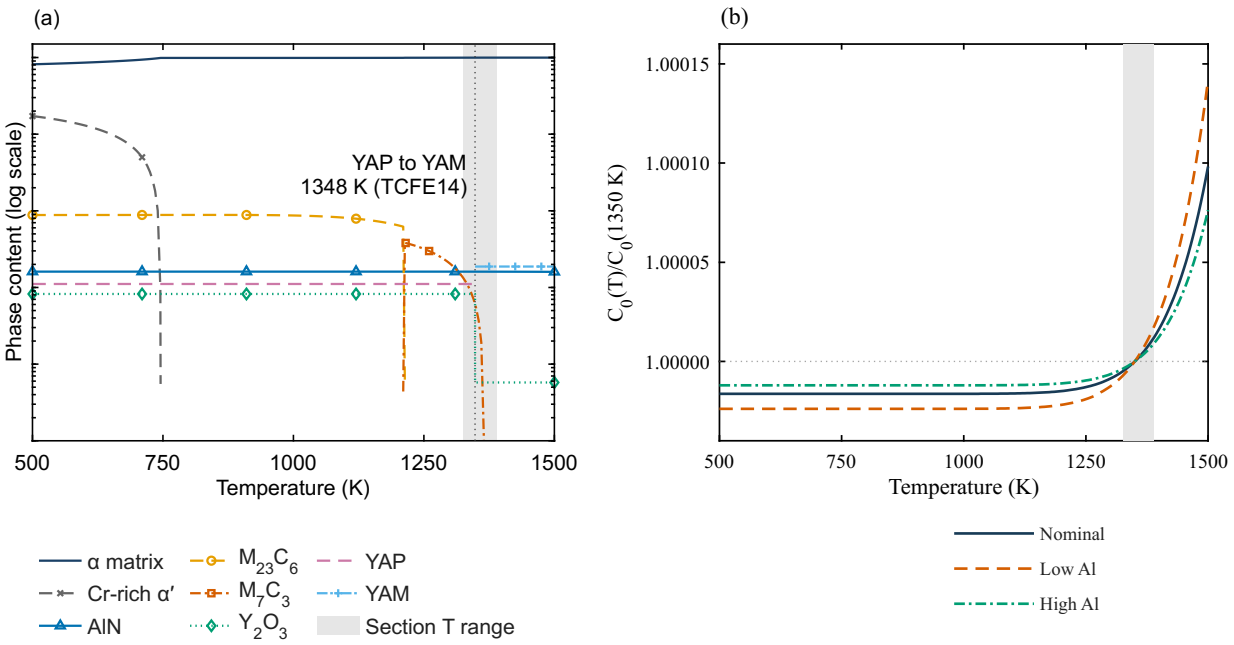


Fig. 4—(a) Calculated equilibrium phase contents of the nominal composition (TCFE14) on a logarithmic axis; the dotted line marks the YAP-to-YAM transition. (b) Available aluminum divided by its value at 1350 K for each composition case. In both panels, shading marks the temperature range of the reference cross-section. Here α is ferrite and α' is the chromium-rich body-centered cubic phase. YAP denotes $YAlO_3$ and YAM denotes $Y_4Al_2O_9$; M in the carbide labels denotes metal.

Table II. Thinning comparisons at T_f = 631 °C and h_{conv} = 62.1 W m^{-2} K^{-1}. Each ratio compares a wall with t_0 at the same exterior coefficient using Eq. 6. T_d is the maximum metal temperature of the cross-section. Activation energies are in kJ mol^{-1}.

h_{ext} (W m^{-2} K^{-1})	d/t_0	T_d (°C)	ΔT (K)	Ratio, Q = 230	Ratio, Q = 360
0	1	1115.8	63.4	1.000	1.000
0	0.814	1100.6	43.1	0.826	0.936
0	0.698	1089.7	28.7	0.713	0.884
20	1	1079.7	38.4	1.000	1.000
20	0.814	1064.7	17.9	0.833	0.949
20	0.698	1054.0	3.3	0.724	0.905

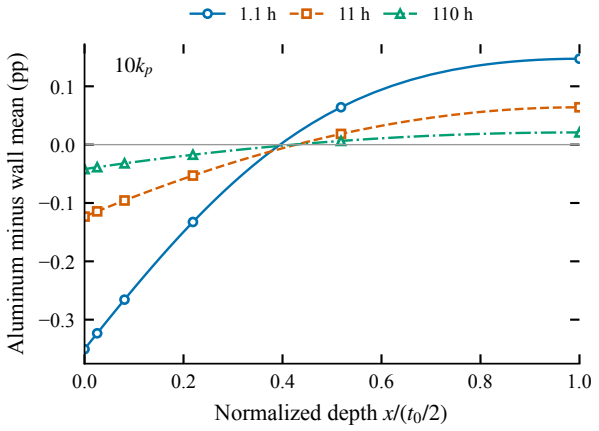


Fig. 5—Calculated aluminum concentration minus the wall mean, in percentage points (pp), across the normalized half-wall depth. In this figure, x is distance from the oxidizing surface, with the wall center at a normalized depth of unity. Curves use $10k_p$ at 1100 °C. Saved times are 1.1, 11 and 110 h.

than the wall mean for k_p and $10k_p$, within the 5% criterion.

IV. DISCUSSION

A. *Temperature field and outer-wall thinning*

Over the tested ranges, channel cooling changed the peak temperature by 187.1 K, and conductivity changed it by 11.9 K. With an insulated exterior, Eq. 3 fixes the mean channel-wall temperature regardless of conductivity. Conductivity therefore mainly set how uneven the temperature was, while channel cooling set the overall level and thus the oxidation rate throughout the section. Interior walls faced channels on both sides, while the outer surface was insulated. This difference in cooling, together with the connected geometry, helps explain the corner maxima.

The thinner walls had shorter depletion times because the loss of aluminum outweighed the benefit of cooling at the tested activation energies. For the insulated walls at 360 kJ mol^{-1}, equal times would have required a further 2.9 K of cooling at $0.814t_0$ and 5.3 K at $0.698t_0$. Exterior convection cooled all three thicknesses by similar amounts. At 20 W m^{-2} K^{-1}, the thinnest section's temperature difference fell from 28.7 to 3.3 K, but its cooling relative to the reference wall changed only from 26.1 to 25.7 K. The ordering of depletion times therefore did not change.

B. *Available aluminum and surface depletion*

The available aluminum stayed nearly constant even though individual phase amounts changed. Naumenko et al.[7] linked carbon–nitrogen interactions with precipitates to scale growth and adherence, a kinetic effect that the equilibrium calculation did not include. In the diffusion calculation, the surface reached the threshold at nearly the same time as the wall mean. The early difference between surface and center therefore did not by itself indicate early loss of protection.

C. *Limitations and further work*

The plate comparisons do not show where the heater would fail first. Local depletion also depends on metal volume per oxidizing area and on aluminum exchange with adjacent walls, which the nominal thickness cannot resolve at corners and junctions.

The calculated times depend on kinetic data from other alloys, specimen thicknesses and exposure times, and on an assumed aluminum threshold. Wright and Peraldi[6] inferred breakaway contents of roughly 1.4–3.8 wt%.

Further work should calculate the depletion time from the local metal volume per oxidizing area. Thickness and temperature would then refer to the same location, so the first depleted region can be found. The calculated hot spots should also be compared with measured metal temperatures and the actual exterior condition of the installed heater. The most important missing input is the activation energy of the heater alloy. For the insulated walls, Eq. 6 gives shorter depletion times for the thinned walls whenever Q is below their respective 429–434 $\mathrm{kJ\,mol^{-1}}$ crossover values.

V. CONCLUSIONS

1. With an insulated exterior, the outer corners were the hottest metal, at 1115.8 °C, 63.4 K above the coolest metal. Channel cooling set the overall temperature, while conductivity mainly set how uneven it was.

2. Oxidation was faster where the metal was hotter: near the outer surface, the parabolic rate constant averaged 1.59–2.03 times the interior value. At equal wall thickness, the hottest corner reached the aluminum threshold in 0.70–0.80 of the time at 1100 °C, and the coolest metal in 2.06–3.10 times that time.

3. Two checks supported the simplified depletion model: available aluminum changed by less than 0.02% over the calculated temperature range, and at 1100 °C the wall surface reached the threshold less than 1% earlier than the wall mean.

4. Thinning the outer wall to 0.698 of its thickness cooled the hottest metal by 26.1 K but shortened depletion time to 0.71–0.88 of the unthinned value, because the thinner wall held less aluminum. Exterior convection up to 20 $\mathrm{W\,m^{-2}\,K^{-1}}$ did not change this. For the insulated walls, thinning would lengthen depletion time only if the activation energy exceeded 429–434 $\mathrm{kJ\,mol^{-1}}$, above both tested values.

ACKNOWLEDGMENTS

The cross-section geometry, the one-dimensional heater model and the alloy certificate were supplied by Sintex A/S.

DECLARATION OF USE OF AI

AI tools assisted with MATLAB data processing, including figure plotting, and with language polishing. The COMSOL thermal simulations and Thermo-Calc calculations were performed independently by the author. The study idea, result analysis and conclusions were developed by the author with the assistance of the two supervisors.

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